Design of Enterprise IoT: Assignment 1b

Smart Turtle Breeding System (STBS)

Team 01

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1 Task 1: Solution Overview

The Smart Turtle Breeding System (STBS) serves 20 indoor or semi-closed Chinese soft-shell turtle units. The architecture separates a device layer, a fog layer and a cloud layer. ESP32-C6 devices collect temperature, pH, ammonia, turbidity, level and flow data; local relay, PWM and driver outputs control the heater, valve and water-cycling pump. A Zigbee coordinator connects the wireless device links and food feeder to a Raspberry Pi 5 edge gateway. The IP camera uses Ethernet through the LAN switch, while the router provides the Internet gateway. The Raspberry Pi aggregates, filters and buffers local data, shows a local dashboard through HDMI, and uploads selected information to the IoT cloud platform over MQTT/HTTPS with TLS. Web, mobile and optional analytics applications support remote access, alerts and longer-term analysis (Munyaev et al., 2026; Prapti et al., 2022; Rastegari et al., 2023).

2 Task 2: IoT Devices

2.1 Overview and suitability

The device set is chosen from the measurements the solution has to make, and it is split so that a fault on one path does not disable the other. ESP32-C6 #1 handles water-environment sensing, while ESP32-C6 #2 checks tank functionality and connects level and flow feedback to the water-change controls. GPIO/ADC, UART/I2C and Pulse/GPIO are used inside the device layer; Zigbee provides the short wireless links to the coordinator. The heater, valve and water-cycling pump remain under local relay, PWM or driver control. The IP camera uses Ethernet and the food feeder uses Zigbee. This arrangement keeps immediate control local while the Raspberry Pi 5 gateway aggregates, filters and buffers data before cloud upload.

2.2 IoT Device Profile

The device profile below identifies the selected field sensors, two ESP32-C6 controllers, actuators, Zigbee food feeder and IP camera. Each row gives the measurable or controllable property, interface, supporting components and design justification. The Raspberry Pi, Zigbee coordinator, switch and router are described in Tasks 3–4.

Table 1: STBS IoT device profile compiled from the official product specifications cited in each row.

Name	Type	Feature of Interest	Property	Characteristics	Components	Justification
ESP32-C6-DevKitC-1 (#1 and #2)	Device	Each breeding tank	Local sensing, actuation and Zigbee link	ESP32-C6-WROOM-1; IEEE 802.15.4 and Zigbee 3.0; GPIO, 12-bit ADC, UART, I^2C, pulse counter and PWM; 3.3 V logic; 8 MB flash. (Espressif Systems, n.d.)	Two development boards, antennas, USB-C/5 V supplies, relay/driver modules, ADC voltage dividers, GPIO level conditioning and RS-485-to-UART interface	Sensing and water-change actuation are split across the two boards so a fault on one path does not disable the other: #1 carries the analogue and RS-485 water-quality channels, #2 carries level, flow and all actuator outputs. The on-die 802.15.4 radio removes the need for an external transceiver, and the ADC, pulse counter, PWM and UART cover every interface used.
DFRobot Waterproof DS18B20 Temperature Sensor (DFR0198)	Sensor	Water in each breeding tank	Water temperature	3.0–5.5 V; -55–125 °C usable range; ±0.5 °C accuracy from -10–85 °C; 9–12-bit resolution; 1-Wire digital interface; waterproof probe. (DFRobot, n.d.-f)	Stainless-steel probe, 1 m cable, pull-up resistor and waterproof cable gland	The 1-Wire signal is a direct GPIO-compatible temperature input for ESP32-C6 #1 and keeps the temperature path simple and digital.
DFRobot Water Quality Ammonia Nitrogen & pH Sensor (SEN0711)	Sensor	Water in each breeding tank	NH_3-N, pH and temperature compensation	NH_3-N 0–100 mg/L; pH 0–14; IP68; 10–30 VDC; RS-485 Modbus-RTU; response time below 30 s; pH and ammonia values are read from the same integrated probe. (DFRobot, n.d.-d)	Integrated probe, 5 m cable, 10–30 VDC supply, DFR0845 isolated RS-485-to-UART converter and shielded twisted pair	One probe reads pH and NH_3-N on the same sample at the same moment, so the two values carry no sampling-point error and one IP68 body serves both. RS-485 Modbus-RTU carries both on one two-wire bus, and the DFR0845 provides the isolated 3.3 V UART interface the ESP32-C6 needs.
DFRobot Gravity Analog Turbidity Sensor (SEN0189)	Sensor	Water in each breeding tank	Relative turbidity	5 VDC; analog 0–4.5 V or digital threshold output; response below 500 ms; probe tip is immersible but the controller board must remain dry. (DFRobot, n.d.-a)	Optical probe, controller board, protective stilling tube, 5 V supply, sealed enclosure and 3.3 V ADC divider	Turbidity is a cheap proxy for suspended solids, with a 0–4.5 V analogue output and a sub-500 ms response. The output is divided to 3.3 V because 4.5 V exceeds the ADC range, and since only the probe tip is immersible the controller board sits in a sealed enclosure.

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Name	Type	Feature of Interest	Property	Characteristics	Components	Justification
DFRobot Gravity Analog Water Pressure Sensor (SEN0257)	Sensor	Water level in each breeding tank	Pressure-derived water level	5 VDC; 0.5–4.5 V linear output; 0–1.6 MPa range; IP68; 0.5–1% full-scale accuracy; response below 2 ms. (DFRobot, n.d.-b)	Stainless pressure sensor, G1/2-to-G1/4 adapter, pressure tap, 5 V supply and 3.3 V ADC divider	It provides an analog level signal for ESP32-C6 #2. The pressure tap can be calibrated to tank depth and the voltage divider protects the controller ADC.
DFRobot Gravity Water Flow Sensor (SEN0217 / YF-S201)	Sensor	Tank inlet and drain pipes	Flow rate and transferred volume	1–30 L/min; Hall-effect pulse output; 3.5–12 VDC; approximately 450 pulses/L; ±5% accuracy over the specified operating range. (DFRobot, n.d.-c)	Rotor, Hall sensor, 1/2-inch fittings, screened cable and 3.3 V-compatible pulse level conditioning	A water change has to be bounded by volume, and about 450 pulses per litre let #2 close the valve on a counted volume rather than on a timer. Inlet and drain are separate branches, so two units per tank also cross-check for a blocked filter or a valve that has not seated.
Aqua Medic Titanium Heater 500 W	Actuator	Water in each breeding tank	Heat input and water temperature	500 W; 110–240 VAC; external controller required; titanium element for aquarium use. (Aqua Medic, n.d.)	Titanium heating element, relay/contactor, RCD, guarded mount, manual isolation and independent overtemperature cut-out	The heater is an electrically powered actuator controlled by a relay or solid-state driver from ESP32-C6 #1. The independent cut-out remains outside the software control loop.
AQUAEL UNI PUMP 1500	Actuator	Tank circulation and water-change line	Water circulation and transfer	220–240 VAC; 19 W; 1400 L/h; maximum head 1.55 m; suitable for submerged or inline operation. (AQUAEL, n.d.)	AC electric pump, strainer, check valve, service bypass, RCD, overload protection and relay/contactor	Rated for continuous duty at 19 W and 1400 L/h, about four tank turnovers an hour through the existing filter at roughly 1 m head, and it runs submerged or inline. Being a 220–240 VAC load it is switched by a rated relay or contactor, with flow feedback confirming that the impeller turns.
Bürkert Type 6213 EV Normally Closed Valve, 24 VDC	Actuator	Tank inlet and drain lines	Water flow state	Servo-assisted 2/2-way diaphragm valve; normally closed; 24 VDC variant; water service; selected variant rated for the pipe pressure and temperature. (Bürkert, n.d.)	Solenoid coil, flyback suppression, 24 VDC supply, manual isolation, upstream strainer and relay/driver	Normally closed action provides a fail-safe boundary. Level and flow feedback allow ESP32-C6 #2 to stop a partial water change when the target or a fault condition is reached.

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Name	Type	Feature of Interest	Property	Characteristics	Components	Justification
Aqara Smart Pet Feeder C1 (PETC1-M01)	Actuator	Dry feed supplied to each breeding unit	Feeding schedule, portion and feeder status	Zigbee IEEE 802.15.4; 5 V, 1 A USB power; 4 L hopper; feeding statistics in grams; local schedules continue without Internet when powered. (Aqara, n.d.)	Hopper, dispensing motor, portion mechanism, USB power cable, Zigbee radio and local schedule memory	A finished Zigbee feeder arrives enclosed and tested, and its stored schedule keeps running without gateway or Internet, so feeding does not stop with the network. Portions are quoted in grams, so pellet size and dispensed mass still need calibrating against turtle feed.
Reolink RLC-811A	Sensor	Turtle activity, feeding and nesting areas	Video and event evidence	8 MP at up to 25 fps; H.264/H.265; 5× optical zoom; PoE IEEE 802.3af or 12 VDC; 10/100 Mbps RJ45; RTSP/HTTPS; IP67; microSD support. (Reolink, n.d.)	Image sensor, lens, processor, PoE/Ethernet interface, mount, privacy mask and local microSD storage	8 MP at 25 fps is 4–8 Mbps, beyond any 802.15.4 link, so this camera must be wired; one PoE cable then carries power (802.3af, under 12 W) and data together, avoiding a mains outlet at each tank. IP67 and the 5× zoom cover a tank from a corner mount, with microSD holding evidence when the uplink fails.

3 Task 3: IoT Network Technology

The device layer is wired by interface requirement rather than by convenience: 1-Wire/GPIO for temperature, RS-485 Modbus through an isolated RS-485-to-UART converter for the combined pH/NH_3-N sensor, GPIO/ADC for the analog turbidity and pressure-derived level sensors, Pulse/GPIO for flow, and relay/PWM/driver outputs for the heater, normally-closed valve and AC electric water-cycling pump (DFRobot, n.d.-e; Espressif Systems, n.d.).

The link between the devices and the network devices is then chosen on range, data rate, latency and power consumption, not on preference for one radio (Rayes & Salam, 2022). Each tank is at most about 30 m by cable route from the coordinator and the whole farm is inside one building, while the aggregated telemetry for all 20 tanks is about 2.4 kbps, so the scenario needs neither long range nor high rate. Zigbee under IEEE 802.15.4 (10–100 m, below 250 kbps, 20–50 ms, very low power) and Wi-Fi (50–100 m, above 100 Mbps, below 10 ms, high power) both cover the distance and both exceed the payload by two orders of magnitude, so range and rate do not separate them. Zigbee is selected because power and spectrum do: its very low power draw lets the radio run from the same 3.3 V rail as the ESP32-C6, and 802.15.4 channels 15, 20 and 25 fall in the gaps between Wi-Fi channels 1, 6 and 11, so the farm Wi-Fi and the device links coexist without contention. Wi-Fi would buy 100 Mbps that 2.4 kbps of telemetry cannot use, in exchange for a DHCP association and a higher idle current at each of

the 20 nodes. Ethernet (100 m per segment, above 1 Gbps, below 1 ms) is rejected for the sensor nodes because cabling 20 tanks is not justified by the payload, but it is kept for the camera, whose 4–8 Mbps at 8 MP and 25 fps is one to two orders of magnitude above the Zigbee ceiling, and for the fog backbone; MQTT/HTTPS over TLS then carries the selected data from the gateway to the cloud. Latency does not discriminate at this scale, because heater, valve and pump decisions are taken by local relay, PWM and driver outputs and never wait for a wireless round trip, so the 20–50 ms Zigbee latency only affects dashboard refresh. 4G is retained as an uplink fallback for the gateway rather than as a device link.

Table 2: STBS IoT network connectivity.

IoT devices, gateways, routers, etc. to connect	**Main considerations**	**Specific questions**	**Responses**
Selected devices: ESP32-C6 #1/#2; field sensors and actuators; Aqara C1 feeder; Reolink camera; Zigbee coordinator; Raspberry Pi 5; monitor; switch; router.	Determine the connectivity between selected IoT devices, gateways, routers, etc.	Why do selected devices need to be connected with each other and to the network?	The water-environment detector connects DFR0198, SEN0711 and SEN0189 to ESP32-C6 #1. The tank-functionality detector connects SEN0257 and SEN0217 to ESP32-C6 #2 and drives the valve and AC electric water-cycling pump. The two ESP32-C6 nodes and Aqara C1 food feeder use the Zigbee path to the coordinator, while the Reolink camera uses Ethernet. The Raspberry Pi 5 aggregates local status, camera data and feeding events before forwarding selected information to the cloud.
	Determine communication and network technologies required to support connectivity	What are the range and coverage requirements?	Each tank is at most about 30 m by cable route from the coordinator and the whole farm sits inside one building, so the requirement is short range with block walls and metal racks in the path. DFR0198 uses 1-Wire/GPIO; SEN0711 uses RS-485 Modbus through the DFR0845 isolated converter to UART; SEN0189 and SEN0257 use analog GPIO/ADC with 3.3 V conditioning; and SEN0217 uses conditioned Hall pulses on GPIO. Zigbee at 2.4 GHz under IEEE 802.15.4 is specified for 10–100 m, which covers the 30 m worst case with margin, against 50–100 m for Wi-Fi and 100 m per segment for Ethernet. The Reolink camera, Raspberry Pi 5, LAN switch and router use Ethernet over RJ45; HDMI connects the local monitor and USB connects the Zigbee dongle.

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IoT devices, gateways, routers, etc. to connect	Main considerations	Specific questions	Responses
		What are the power consumption constraints?	The power budget falls into three classes. The ESP32-C6 nodes, sensors, coordinator and Aqara feeder run from low-voltage local supplies, and Zigbee is classed as very low power, whereas the Wi-Fi alternative is classed as high power; that difference, not range, is why the 20 nodes do not use Wi-Fi, because 20 mains-fed stations would still add 20 associations and 20 idle draws to the site AP. The valve uses a 24 VDC supply; the heater and the water-cycling pump are mains-powered and are the two dominant loads; and the Reolink camera draws under 12 W over PoE. The LAN switch, Raspberry Pi 5, monitor and router require suitable local power. RCD/overload protection and a local backup should preserve the gateway, coordinator and controllers long enough to record alarms and keep bounded local control available.
		What is the required data rate (how much data needs to be transmitted over time)?	The measured payload is small: 15 records per tank per minute at about 60 bytes each is roughly 2.4 kbps for all 20 tanks before MQTT and TLS overhead, which is two orders of magnitude below the 250 kbps IEEE 802.15.4 ceiling and four below the Wi-Fi figure, so telemetry alone does not separate Zigbee from Wi-Fi. The IP camera is the one bandwidth-heavy device: 8 MP at 25 fps needs 4–8 Mbps, above the Zigbee ceiling, and therefore uses Ethernet to the LAN switch. The Raspberry Pi 5 aggregates and filters the local streams, so the cloud receives selected telemetry, events or processed video rather than requiring every raw local signal to travel continuously over the Internet.
		What are the latency requirements (how fast does data need to be transmitted)?	Heater, valve and AC pump decisions use local relay, PWM or driver signals and do not wait for the cloud. Sensor feedback reaches the ESP32-C6 controllers over 1-Wire/GPIO, RS-485/UART, GPIO/ADC or Pulse/GPIO links. Zigbee and Ethernet support gateway monitoring and event delivery, while MQTT/HTTPS cloud communication is used for reporting and alerts rather than immediate safety action.

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IoT devices, gateways, routers, etc. to connect	Main considerations	Specific questions	Responses
		What are the security and reliability requirements?	Local control should continue when the router or Internet is unavailable. The Raspberry Pi 5 stores data locally when upload is interrupted and the gateway labels or forwards buffered events after reconnection. MQTT/HTTPS traffic to the cloud uses TLS; Zigbee devices require authenticated joining and coordinator access control; and actuator commands require local bounds and interlocks. RS-485 isolation, ADC protection, GPIO level conditioning, RCD/overload protection, cable glands and waterproof enclosures are required.
		What are the cost considerations?	ESP32-C6 nodes and short Zigbee links reduce point-to-point wiring inside the device layer. Ethernet is reserved for the Reolink IP camera and the fixed fog/network path, while one Raspberry Pi 5 provides local aggregation, buffering and display output. The selected sensors require signal conditioning and protected enclosures; the AC pump and heater require correctly rated switching and electrical protection. Cloud storage and analytics can be scaled according to selected uploads.
		Which communication/network technologies are selected?	Field interfaces are 1-Wire/GPIO for temperature; RS-485 Modbus through DFR0845 and UART for the combined pH/NH3-N sensor; GPIO/ADC for turbidity and pressure-derived level; Pulse/GPIO for flow; and Relay/PWM/Driver for heater, valve and AC pump control. For the links between devices and network devices the decision is Zigbee at 2.4 GHz under IEEE 802.15.4 for the ESP32-C6 and Aqara feeder, Ethernet at 10/100/1000 Mbps over RJ45 for the Reolink camera and the fog/network path, USB for the coordinator, HDMI for the local monitor, and MQTT/HTTPS over TLS to the cloud platform. Zigbee wins over Wi-Fi because range and rate are satisfied by both while power and channel sharing are not: very low power against high, and 802.15.4 channels 15/20/25 sit in the gaps between Wi-Fi channels 1/6/11 so the two coexist. Ethernet is kept only where the payload or the fixed path justifies cable, and 4G is held as an uplink fallback rather than a device link.

4 Task 4: Fog and Cloud Computing

The Raspberry Pi 5 is the fog gateway for data aggregation, local control rules, filtering, optional video processing, local buffering and upload to the cloud. The USB Zigbee coordinator bridges the ESP32-C6 and Aqara feeder links to the gateway. The HDMI-connected local monitor is display-only. The router provides the Internet gateway, while the IoT cloud platform supports long-term storage, analytics, notifications, remote access, model training and device management. The cloud is therefore useful for applications and historical analysis but is not placed in the immediate actuator control loop (Raspberry Pi, n.d.; SONOFF, n.d.).

Table 3: STBS fog and cloud connectivity.

IoT devices and network devices (e.g., router, hub, gateway, etc.)	**Main considerations**	**Specific questions**	**Responses**
Edge/fog devices: ESP32-C6 #1/#2; Aqara C1 feeder; SONOFF ZBDongle-E coordinator; Raspberry Pi 5; monitor; switch; router.	Determine the connectivity between IoT devices and network devices in the edge/fog layer	Is an edge/fog layer required? Why?	Yes, because two requirements cannot be met from the cloud: local control has to survive an Internet outage, and 20 continuous video streams are too costly in bandwidth and too sensitive in privacy to send out routinely. A fog layer between the device layer and the cloud follows from those two requirements. The Raspberry Pi 5 provides data aggregation, local control rules, data filtering, optional video processing, local buffering and upload to the cloud. This keeps the system useful when the Internet is unavailable and prevents every local signal from depending on a cloud round trip. The SONOFF coordinator bridges the ESP32-C6 and Aqara Zigbee links to the gateway.
		Which network device(s) should handle computing across the edge/fog layer?	The Raspberry Pi 5 is the edge gateway and performs the fog-layer computing. The SONOFF ZBDongle-E acts as the Zigbee 3.0 USB coordinator, the LAN switch forwards Ethernet traffic, the router provides the Internet gateway, and the HDMI-connected local monitor is display-only.
		What data is transmitted between the IoT devices and network device(s)?	The ESP32-C6 nodes send temperature, pH, NH3-N, turbidity, water level and flow readings, together with actuator states and food-feeder events. The IP camera sends monitoring video over Ethernet. The Raspberry Pi 5 adds timestamps, device identity, local status and event context before forwarding selected data to the cloud.

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IoT devices and network devices (e.g., router, hub, gateway, etc.)	Main considerations	Specific questions	Responses
		What data processing and/or analysis tasks should be performed in the edge/fog layer?	The gateway aggregates sensor readings, applies local control rules, filters noisy or unnecessary data, buffers records during Internet loss and can perform optional video processing. It also correlates sensor feedback with heater, valve, pump and feeder actions so the local dashboard can show whether a commanded operation has produced the expected result.
		If the fog connects with a user application, what is the purpose of the connection?	The local monitor receives an HDMI display-only connection from the Raspberry Pi 5 and presents a local dashboard. The dashboard shows current water conditions, actuator states, feeder status, camera or event information and gateway health without requiring the cloud connection to be available.
		If the fog connects with a system, what is the purpose of the connection?	The Raspberry Pi 5 connects through the LAN switch and router to the IoT cloud platform. It uploads selected telemetry, events, alerts and optional processed video over MQTT/HTTPS with TLS. The gateway can store data locally and forward buffered records after the Internet connection returns.
Cloud path: Raspberry Pi 5; SONOFF coordinator; router; IoT cloud platform; web dashboard; mobile app; optional analytics.	Determine the connectivity to the cloud	Is a cloud layer required? Why?	A cloud layer is required for the tasks that cannot be held on one local gateway: long-term data storage, analytics and visualisation, alerts, remote access, model training and device management, together with multi-user access and history that survive the loss of the site. It is not required for the immediate relay, PWM or driver control in the device layer, which is why it is kept off the control path.
		What data should be transmitted to the cloud, and is it from IoT devices or edge/fog layer devices?	The Raspberry Pi 5 should publish aggregated sensor readings, actuator and feeder states, alarms, gateway health and selected event records. Camera data can be uploaded as selected or processed evidence when the optional analytics function is enabled. Raw local control signals remain at the device and fog layers.
		What data processing and/or analysis tasks should be performed in the cloud?	The cloud stores long-term records, provides data analytics and visualisation, issues alert and notification workflows, supports remote access through web or mobile applications, enables behaviour or nesting model training when required, and manages authorised device configuration.

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IoT devices and network devices (e.g., router, hub, gateway, etc.)	Main considerations	Specific questions	Responses
		If the cloud connects with a user application, what is the purpose of the connection?	The web dashboard provides farm operators with remote status, historical trends, event records and authorised monitoring functions. The mobile app delivers alerts and notifications so operators can respond when a condition requires attention.
		If the cloud connects with a system, what is the purpose of the connection?	The optional data-analytics service receives approved cloud data for longer-term comparison, behaviour or nesting analysis and reporting. Cloud-originated configuration requests must be authenticated and bounded by the local gateway and ESP32-C6 control logic before they can affect an actuator.

5 Task 5: Overall Solution Connectivity

5.1 Graphical representation

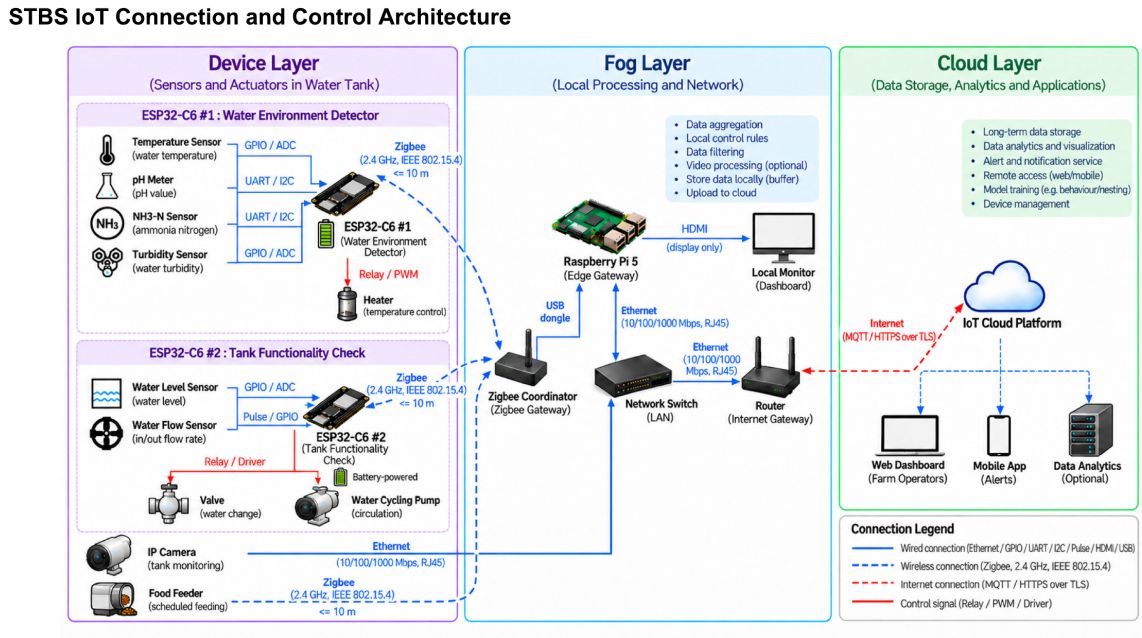


Figure 1: STBS connectivity and control architecture.

5.2 Description of the diagram

Figure 1 follows the three-layer structure and connection legend: the device layer senses and actuates, the fog layer aggregates, filters and buffers local data, and the cloud layer stores, analyses and presents authorised results. ESP32-C6 #1 and #2 implement the two local device functions, with 1-Wire/GPIO, RS-485/UART, GPIO/ADC and Pulse/GPIO connections to the sensors and relay/PWM/driver outputs to the heater, valve and AC electric pump. The Aqara food feeder is a native Zigbee device; the IP camera uses Ethernet RJ45. The Raspberry Pi 5, USB Zigbee coordinator, switch and router provide the fog/network path, while MQTT/HTTPS over TLS carries selected data to the cloud. Immediate control remains local and does not depend on a cloud round trip.

6 Task 6: Conclusion

The design links continuous water and feed measurements, bounded actuation, video evidence and role-specific alerts across 20 units. Immediate sensing and actuation stay in the device layer, Zigbee and Ethernet connect the local system, and aggregation, filtering and buffering are assigned to the Raspberry Pi 5 fog gateway. The water-cycling pump is an AC electric actuator, while the food feeder is a native Zigbee device. The cloud adds long-term storage, visualisation, alerts, remote access and optional behaviour or nesting analytics. The main limitations are probe fouling and calibration effort, Zigbee range and 2.4 GHz contention over the 30 m worst-case links, dependence on local gateway and switch power, and the need to validate video models under changing light, water and turtle density. Ammonia and video inference are indicators rather than diagnoses, while feeder runtime only estimates delivered mass. A staged pilot should validate thresholds, wireless coverage, hydraulic timings, electrical protection, false-alarm rates, model performance and operating procedures before all 20 units are commissioned.

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